

COMPUTER FUNDAMENTALS NOTES

MS EXCEL BASICS, ALGORITHMS & FLOWCHARTS

Chapter 1: Microsoft Excel Basics

What is Microsoft Excel?

Microsoft Excel is a spreadsheet application developed by Microsoft that is used to organize, store, calculate, and analyze data.

Excel helps users perform calculations quickly and efficiently using formulas and functions.

Common Uses of Excel

- Storing student records
 - Calculating marks
 - Managing expenses
 - Creating attendance sheets
 - Generating reports
 - Analyzing data
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Main Components of Excel

Workbook

A workbook is an Excel file that contains one or more worksheets.

Worksheet

A worksheet is a page inside a workbook where data is entered and managed.

Rows

Rows run horizontally and are identified by numbers.

Example: 1, 2, 3, 4, 5...

Columns

Columns run vertically and are identified by letters.

Example: A, B, C, D, E...

Cell

A cell is the intersection of a row and a column.

Example: A1, B2, C5

Entering Data in Excel

Different types of data can be entered into Excel:

Text Data

Examples:

- Bharath
- Student
- Computer

Number Data

Examples:

- 10
- 45
- 98

Date Data

Examples:

- 01/01/2025
 - 15/08/2025
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Important Excel Functions

Functions are predefined formulas that help perform calculations automatically.

1. SUM Function

Used to add numbers.

Formula:

`=SUM(A1:A5)`

Example:

$10 + 20 + 30 + 40 + 50 = 150$

Result:

150

2. AVERAGE Function

Used to calculate the average of numbers.

Formula:

`=AVERAGE(A1:A5)`

Example:

Marks: 80, 90, 70

Average:

$(80 + 90 + 70) \div 3 = 80$

Result:

80

3. MAX Function

Used to find the largest value.

Formula:

`=MAX(A1:A5)`

Example:

10, 25, 90, 40

Result:

90

4. MIN Function

Used to find the smallest value.

Formula:

=MIN(A1:A5)

Example:

10, 25, 90, 40

Result:

10

5. COUNT Function

Used to count the number of cells containing numbers.

Formula:

=COUNT(A1:A5)

Example:

10, 20, 30, 40

Result:

4

Sample Student Marks Table

Name	Maths	Science	English
Rahul	80	90	70
Aryan	75	85	90
Riya	95	88	92

Functions can be used to calculate:

- Total Marks
 - Average Marks
 - Highest Marks
 - Lowest Marks
-

Chapter 2: Algorithms

What is an Algorithm?

An algorithm is a step-by-step procedure used to solve a problem or complete a task.

Algorithms help computers understand exactly what needs to be done.

Characteristics of a Good Algorithm

Clear

Every step should be easy to understand.

Ordered

Steps should be arranged in the correct sequence.

Finite

The algorithm must end after a certain number of steps.

Effective

Each step should help solve the problem.

Everyday Examples of Algorithms

Algorithm for Brushing Teeth

Step 1: Take a toothbrush

Step 2: Apply toothpaste

Step 3: Brush teeth

Step 4: Rinse mouth

Step 5: Finish

Algorithm for Making Tea

Step 1: Boil water

Step 2: Add tea powder

Step 3: Add sugar

Step 4: Add milk

Step 5: Stir and serve

Algorithm for Logging into Gmail

Step 1: Open a web browser

Step 2: Go to Gmail

Step 3: Enter email address

Step 4: Enter password

Step 5: Click Sign In

Step 6: Access mailbox

Importance of Algorithms

- Helps solve problems systematically
 - Makes programming easier
 - Improves logical thinking
 - Reduces mistakes
 - Saves time
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Chapter 3: Flowcharts

What is a Flowchart?

A flowchart is a graphical representation of an algorithm.

It uses different symbols to show the sequence of steps required to solve a problem.

Flowcharts help programmers and students understand processes easily.

Components of a Flowchart

1. Start / End Symbol (Oval)

Purpose: Represents the beginning or end of a process.

Example:

START

END

2. Process Symbol (Rectangle)

Purpose: Represents an action or task.

Examples:

- Add two numbers
 - Open browser
 - Enter name
-

3. Decision Symbol (Diamond)

Purpose: Represents a condition or question.

Examples:

- Is it raining?
- Is marks greater than 40?
- Is password correct?

Possible outcomes:

- Yes
 - No
-

4. Input / Output Symbol (Parallelogram)

Purpose: Used to take input or display output.

Examples:

- Enter marks
 - Display result
-

5. Flow Line (Arrow)

Purpose: Shows the direction of the process.

It connects all symbols together.

Example Flowchart: Student Pass or Fail

START



Enter Marks



Is Marks $\geq$ 40?

↙ ↘

YES NO

↓ ↓

PASS FAIL

↓ ↓

END END

Example Flowchart: Taking an Umbrella

START

↓

Is it raining?

↙ ↘

YES NO

↓ ↓

Take Don't Take

Umbrella Umbrella

↓ ↓

END END

Advantages of Flowcharts

- Easy to understand
- Improves problem-solving skills
- Helps identify mistakes
- Makes programming easier
- Provides visual representation of a process

Quick Revision

1. What is Microsoft Excel?
 2. What is a workbook?
 3. What is a worksheet?
 4. What does the SUM function do?
 5. What does the AVERAGE function do?
 6. What is an algorithm?
 7. Why are algorithms important?
 8. What is a flowchart?
 9. Name the symbols used in a flowchart.
 10. What is the purpose of a decision symbol?
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Homework

1. Create an Excel sheet containing marks of five students and calculate:
 2. Total Marks
 3. Average Marks
 4. Highest Marks
 5. Lowest Marks
6. Write an algorithm for:
 7. Going to School
 8. Making Maggi
9. Draw flowcharts for:
 10. Student Pass or Fail
 11. Checking Whether It Is Raining